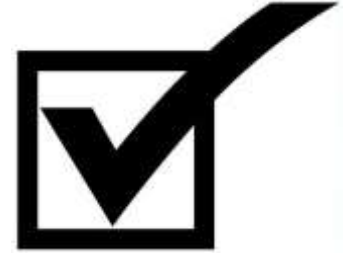




**BUSINESS
PROFESSIONALS
of AMERICA**
Giving Purpose to Potential



MANAGERIAL ACCOUNTING (135) NATIONAL 2023

CONCEPT KNOWLEDGE:

Multiple Choice (20 @ 2 points each) _____ (40 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (54 points)

Problem 1: Budgeted Income Statement _____ (51 points)

Problem 2: Flexible Budgets _____ (65 points)

Problem 3: Capital Budgets _____ (30 points)

TOTAL POINTS _____ (***240 points***)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Multiple Choice

Identify the letter of the choice that *best* completes the statement or answers the question.

1. C
2. B
3. D
4. D
5. D
6. C
7. A
8. B
9. C
10. A
11. B
12. C
13. D
14. D
15. D
16. C
17. B
18. A
19. A
20. A

Short Answer (54 Points; each answer is worth 3 points)

1. Renner company makes and sells dog harnesses. Each harness uses $\frac{3}{4}$ yard of nylon. Budgeted production of harnesses in units for the next three months is as follows:

	June	July	August
Budgeted production	21,000	20,000	24,000

The company wants to maintain monthly ending inventories of plastic equal to 25% of the following month's budgeted production needs; beginning inventory is the expected amount. The cost of nylon is \$2.12 per yard.

How many yards of nylon should Renner purchase in June?	\$15,562.50
What is the budgeted cost of materials for June?	\$32,992.50

2. Barton Inc. provided the following information:

	March	April	May
Projected merchandise purchases	\$76,000	\$65,000	\$70,000

Barton pays for 40% of merchandise purchases in the month purchased. General operating expenses are budgeted to be \$20,000 per month, of which \$2,000 is depreciation. Barton pays operating expenses in the month incurred.

What are Barton's cash disbursements for May?	\$85,000
What will be the balance in Accounts Payable at the end of May?	\$42,000

3. Cardellini Inc.'s manufacturing costs for May when production was 1,000 units appears below:

Direct materials	\$12,000
Direct labor	6,500
Variable overhead	5,000
Factory depreciation	9,000
Factory supervisor salaries	7,800
Other fixed factory costs	2,500

What is the flexible budget manufacturing cost for a month when 800 units are produced?	\$38,100
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4. Steinfeld Company combines its operating expenses for budget purposes in a selling and administrative expense budget. For the first quarter, the following data are compiled:

Sales: 20,000 units; unit selling price:	\$35
Variable costs per dollar of sales:	
Sales commissions	6%
Delivery expense	2%
Advertising	4%
Fixed costs per quarter:	
Sales salaries	\$24,000
Office salaries	17,000
Depreciation	6,000
Insurance	2,000
Property taxes	1,000

What is the selling and administrative expense budget for the first quarter?	\$134,000
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5. Whitfield Company produced and sold 50,000 units of its product and is operating at 70% of plant capacity. Each unit sells for \$70, and costs \$55 to produce. Fixed costs are 18% of the total manufacturing costs. Whitfield has received a proposal from another company to buy 15,000 units for \$50 per unit.

Should Whitfield accept the special order? Yes or no.	Yes
What effect would acceptance of this order have on the company's income?	\$73,500

6. Bartlet's Bikes has been manufacturing its own seats for its bicycles. The company is currently operating at 100% capacity, and variable manufacturing overhead is charged to production at the rate of 60% of direct labor cost. The direct materials and direct labor cost per unit to make the bicycle seats are \$8.00 and \$9.00, respectively. Normal production is 50,000 bicycles per year. A supplier offers to make the bicycle seats for \$22 each. If Bartlet accepts this offer, all variable manufacturing costs will be eliminated, but the \$30,000 of fixed manufacturing overhead currently being charged to the bicycle seats will have to be absorbed by other products.

Should Bartlet buy the seats from the outside supplier? Yes or no.	Yes
What effect would buying the seats from the outside supplier have on the company's income?	\$20,000

7. Olivia is the manager of Cat Castles, which manufactures climbing condos for cats. Cat Castles sells their condos to retailers unassembled for \$400. The cost of an unassembled condo is:

Direct materials	\$150
Direct labor	70
Variable overhead (70% of direct labor)	49
Fixed overhead (30% of direct labor)	<u>21</u>
Manufacturing cost per unit	<u>\$290</u>

Olivia has determined that their unused capacity is expected to continue indefinitely, and that some of that capacity can be used to assemble the condos and sell them for \$440. Assembling the condos will increase direct materials by \$5 per unit, and direct labor by \$10 per unit. Additional variable overhead will be incurred at the normal rates, but there is not expected to be an additional fixed overhead as a result of assembling the condos.

Should Cat Castles sell the condos assembled or unassembled?	Assembled
What effect would selling the condos assembled have on the company's income?	\$18 per unit

8. Scantling Co. relies heavily on its 3D printer to print models of its products. The current printer is an older model and is inefficient. Management is considering updating the printer with a new model.

	Current Printer	New Model
Original purchase cost	\$10,000	\$20,000
Accumulated depreciation	8,000	
Estimated operating costs (annual)	9,000	4,200
Remaining useful life	5 years	5 years

If sold now, the current printer would have a salvage value of \$1,000; if operated for the remainder of its useful life, the salvage value is estimated to be zero. The new machine is expected to have a zero salvage value after five years.

Should Scantling buy the new machine or keep the old machine?	Buy new
What effect would buying the new machine have on income?	Save \$5,000

9. Rodgers Inc. has three divisions: Green, Gold, White. The results of the first quarter of 2022 are presented below.

	Green	Gold	White	Total
Units sold	3,000	5,000	2,000	10,000
Revenue	\$70,000	\$50,000	\$40,000	\$160,000
Variable costs	32,000	26,000	16,000	74,000
Fixed costs	<u>20,000</u>	<u>32,000</u>	<u>16,000</u>	<u>68,000</u>
Net income	<u>\$18,000</u>	<u>(\$8,000)</u>	<u>\$8,000</u>	<u>\$18,000</u>

Allocated fixed costs are 43.75% of total fixed costs. Rodgers estimates that all the allocated costs will continue even if a division is discontinued. Since the Gold division is operating at a loss, Rodgers feels that it should be discontinued. If the Gold division is closed, it is estimated that sales of the Green division will increase by 10%, and sales at the White division will remain unchanged.

Should Rodgers discontinue the Gold division? Yes or no.	No
What effect would discontinuing the Gold division have on the company's income?	(\$2,200)

10. Crosby Company manufactures and sells two products. Relevant per unit data concerning each product is below.

	Standard	Deluxe
Selling price	\$28	\$32
Variable costs	\$10	\$12
Machine hours	4	5

If 1,000 additional machine hours are available, which product should be manufactured?	Standard
If all the additional machine hours are allocated to the product you identified above, what would be the effect on total contribution margin?	\$4,500

Problem 1 (51 Points; each line of the heading, each account title, and each amount worth 3 points)

In September, the budget committee of Bishop Company assembles the following data:

- Expected Sales
 - October \$900,000
 - November \$850,000
 - December \$800,000
- Cost of goods sold is expected to be 60% of sales
- Desired ending inventory is 20% of next month's cost of goods sold
- The beginning inventory on October 1 will be the desired amount

Prepare the budgeted income statement for October through gross profit, including a cost of goods sold schedule.

Bishop Company		
Income Statement		
Month Ended October 31		
Sales		90,000
Beginning Inventory	108,000	
Purchases	<u>534,000</u>	
Goods Available for Sale	642,000	
Ending Inventory	<u>(102,000)</u>	
Cost of Goods Sold		<u>(540,000)</u>
Gross Profit		360,000

Problem 2 (65 Points; each shaded box in the budget report worth 3 points; each variance answer is worth 5 points. Answers must include F or U designation to be correct)

Lyman Company produces men's ties. Each tie is budgeted to use 1.5 yards of silk and 45 minutes of direct labor. In 2022, Lyman purchased 11,600 yards of silk and used 9,860, and Lyman's tailors worked a total of 4,220 hours. The following budgeted and actual amounts are for 2022:

	Budget at 5,000 units	Actual amounts at 5,800 units
Direct materials	\$60,000	\$70,992
Direct labor	75,000	86,510
Equipment depreciation (straight-line)	5,000	5,000
Indirect labor	7,500	8,600
Indirect materials	9,000	9,600
Rent & insurance	12,000	13,000

Prepare a budget report for Lyman Company for the year.

Grader: Each shaded box is worth 3 points

	Flexible Budget	Actual	Difference
Direct Materials	\$69,600	\$70,992	\$1,392 U
Direct Labor	87,000	86,510	490 F
Depreciation	5,000	5,000	0
Indirect Labor	8,700	8,600	100 F
Indirect Materials	10,440	9,600	840 F
Rent & Insurance	<u>12,000</u>	<u>13,000</u>	<u>1,000 U</u>
Total	\$192,740	\$193,702	\$962 U

Grader: Each answer below is worth 5 points

What is the direct materials price variance?	\$21,808 F
What is the direct materials usage variance?	\$9,280 U
What is the direct labor rate variance?	\$2,110 U
What is the direct labor efficiency variance?	\$2,600 F

Problem 3 (30 Points; each answer worth 5 points)

Valdez Company is considering purchasing a new summer camp in the mountains of Wisconsin for \$3,000,000. The new facility will generate annual net cash inflows of \$500,000 for ten years. Specialists estimate that the facility will remain useful for ten years and have no residual value. The company uses straight-line depreciation, and its stockholders demand an annual return of 10% on investments of this nature.

Calculate the payback period of this investment	6 years
Calculate the Accounting Rate of Return of this investment	13.3%
Calculate the Net Present Value of this investment	\$72,500
Calculate the approximate Internal Rate of Return of this investment	11% (Grader: accept any answer between 10% and 12%)
Calculate the Profitability Index of this investment	.02
Should Valdez invest in this project? Yes or no.	Yes